MAIS PAC

IPhO 02.14: Structure of Matter and Nuclear Physics · Worked Solutions

Structure of Matter and Nuclear Physics: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

1 2 3 4 5 6 7 8 9 10

Solution 1 — Nuclear notation. Return →

(a) Z = 92 protons, A - Z = 235 - 92 = 143 neutrons, and A = 235 nucleons.

(b) m = 235.04 {u}×1.661×10^(-27) {kg,u^(-1)} = 3.90×10^(-25) {kg}.

(c) Isotopes have the same number of protons but different numbers of neutrons, so the chemistry is identical and only the mass and nuclear stability differ. Uranium-235 and uranium-238 are isotopes, with half-lives of 7×10^8 years and 4.5×10^9 years respectively. **Check:** 235×931.5 {MeV}/c² gives a rest energy of about 219 {GeV}, consistent with mc² = (3.90×10^(-25))(9×10^(16)) = 3.5×10^(-8) {J}.

Solution 2 — Balancing nuclear equations. Return →

(a) A: 238 - 234 = 4, Z: 92 - 90 = 2, so the product is ^4_2{He} — **alpha decay**.

(b) A unchanged at 14, Z increases by 1 and charge must balance, so the product is e^- + _e — **beta minus decay** (an electron and an antineutrino).

(c) A: 2+3-4 = 1, Z: 1+1-2 = 0, so the product is a neutron, ^1_0{n} — **fusion**, releasing 17.6 {MeV}. **Check:** in each case the totals of A and Z on both sides match, which is the only test needed.

Solution 3 — Mass defect of helium-4. Return →

(a) Δ m = 2(1.00728) + 2(1.00866) - 4.0015 = 4.03188 - 4.0015 = 0.03038 {u}.

(b) E_b = 0.03038×931.5 = 28.3 {MeV}, so E_b/A = 7.08 {MeV} per nucleon — unusually high for such a light nucleus.

(c) The alpha particle has a closed shell of two protons and two neutrons, which is a particularly stable configuration. That is why heavy nuclei emit alphas rather than loose nucleons: the alpha carries away a large amount of energy for a small mass, and the daughter nucleus moves closer to stability. **Check:** the mass defect is about 0.75% of the total mass, comparable with the 0.4% figure quoted for fusion.

Solution 4 — Half-life and activity. Return →

(a) λ = (ln 2)/(T_(1/2)) = (0.693)/(8.0×86,400 {s}) = 1.00×10^(-6) {s^(-1)}.

(b) A_0 = λ N_0 = (1.00×10^(-6))(4.0×10^(15)) = 4.0×10^9 {Bq} = 4.0 {GBq}.

(c) 24 days is three half-lives, so A = A_0/8 = 5.0×10^8 {Bq} and N = 5.0×10^(14). **Check:** A = λ N = (1.00×10^(-6))(5.0×10^(14)) = 5.0×10^8, consistent.

Solution 5 — Carbon dating. Return →

(a) Activity is proportional to the number of ^(14){C} nuclei, so

$$(A)/(A_0) = e^{(-\lambda t)}\ t = (1)/(\lambda)\ln ()/(){A_0}{A} = (5730)/(0.693)\ln ()/(){2.3}{1.5} = 8267\times 0.427 = 3.5\times 10^{3}\ \{\text{years}\}. \tag{1}$$

(b) The method assumes that the ^(14){C} concentration in the atmosphere has been constant (it varies with solar activity and with burning of fossil fuel), that the sample has not exchanged carbon with its surroundings since death, and that no contamination has occurred.

(c) After 50,000 years (about 8.7 half-lives) only 0.2% of the original ^(14){C} remains, so the activity is close to background and the uncertainty becomes enormous. **Check:** at 3500 years the fraction remaining is e^(-0.693×3500/5730) = 0.65, and 1.5/2.3 = 0.65, as it must.

Solution 6 — Energy from fission. Return →

(a) E = 200×10^6(1.60×10^(-19)) = 3.2×10^(-11) {J} per fission.

(b) 1.0 {kg} of ^(235){U} contains N = (1000)/(235)(6.02×10^(23)) = 2.56×10^(24) nuclei, so

$$E = (2.56\times10^{24})(3.2\times10^{-11}) = 8.2\times10^{13}\ \{J\}. \tag{2}$$

(c) Δ m = E/c² = 8.2×10^(13)/(9.0×10^(16)) = 9.1×10^(-4) {kg}, which is 0.09% of a kilogram. **Check:** the same figure follows from 200 {MeV} per 235 {u}: 200/931.5×235 = 0.09%.

Solution 7 — Alpha particle energy. Return →

(a) K = 5.5 {MeV} = 8.8×10^(-13) {J} with m_α = 6.64×10^(-27) {kg}: v = √(2K/m) = √(2(8.8×10^(-13))/(6.64×10^(-27))) = 1.6×10^7 {m,s^(-1)}, about 5% of c.

(b) Momentum conservation: p_α = m_α v = 1.08×10^(-19) {kg,m,s^(-1)}, so the recoiling nucleus has v = p/M = 1.08×10^(-19)/(220×1.66×10^(-27)) = 2.96×10^5 {m,s^(-1)}.

(c) The alpha is heavy and doubly charged, so it ionises matter very densely and loses energy within a few centimetres of air or a sheet of paper. A gamma ray is uncharged and interacts only by occasional collisions, so it penetrates deeply — which is also why it is the most penetrating hazard. **Check:** the recoil kinetic energy, about 0.1 {MeV}, is a small fraction of the alpha energy, as momentum conservation with a heavy daughter requires.

Solution 8 — The binding energy curve. Return →

(a) E ≈ A[(B/A)_({fragments}) - (B/A)_({parent})] = 238(8.5 - 7.6) = 2.1×10² {MeV}.

(b) This is very close to the observed 200 {MeV}: the estimate only uses the average binding energy per nucleon, and the agreement to about 5% shows how much of nuclear physics follows from the curve alone.

(c) Per kilogram, fusion wins because the fractional mass change is larger: hydrogen converts about 0.7% of its mass in the pp chain, whereas fission converts about 0.09%. The Sun fuses 4×10^9 {kg} of mass per second for this reason. **Check:** the 0.7% figure is 4×(7.07 {MeV})/(4×938 {MeV}), using the helium binding energy from Problem 3.

Solution 9 — A sealed source. Return →

(a) N_0 = (2.0×10^(-3))/(60)(6.02×10^(23)) = 2.0×10^(19) atoms.

(b) λ = (0.693)/(5.3×3.16×10^7) = 4.1×10^(-9) {s^(-1)}, so

$$A_0 = \lambda N_0 = (4.1\times10^{-9})(2.0\times10^{19}) = 8.3\times10^{10}\ \{\mathrm{Bq}\}. \tag{3}$$

(c) 20 years is 3.77 half-lives, so A = A_0,2^(-3.77) = 6.1×10^9 {Bq}. The nuclei remaining are N = N_0,2^(-3.77) = 1.5×10^(18), so the number of decays is N_0 - N = 1.85×10^(19). **Check:** the total energy released is then about 1.85×10^(19)×{(a few MeV)} ≈ 10 {GJ}, and the **power** starts at 8.3×10^(10) decays per second — a useful reminder that a source with a long half-life can still be intensely radioactive.

Solution 10 — A nuclear reactor. Return →

(a) 1.0 {GW} = 1.0×10^9 {J,s^(-1)} of thermal power, and each fission gives 3.2×10^(-11) {J}, so

$$\{\text{rate}\} = (1.0\times10^{9})/(3.2\times10^{-11}) = 3.1\times10^{19}\ \{\text{fissions per second}\}. \tag{4}$$

Per day the mass consumed is ((3.1×10^(19))(86,400))/(6.02×10^(23))×235 {g} = 1.05×10³ {g} ≈ 1.0 {kg} of ^(235){U}.

(b) P_({electrical}) = 0.33×1.0 {GW} = 3.3×10^8 {W}, so 330 {MW}.

(c) Natural uranium is only 0.7% ^(235){U}, and a reactor needs a higher proportion to sustain a chain reaction with ordinary (light) water, so the fuel is enriched. The moderator slows the fast neutrons produced in fission to thermal energies, where the fission cross-section of ^(235){U} is far larger — water, heavy water and graphite are the usual moderators. **Check:** 1 {kg} of ^(235){U} is about 8×10^(13) {J}, so a 1 {GW} reactor burning a kilogram a day produces 8×10^(13) {J} per day, i.e. 9.5×10^8 {J,s^(-1)} — consistent with the thermal power.